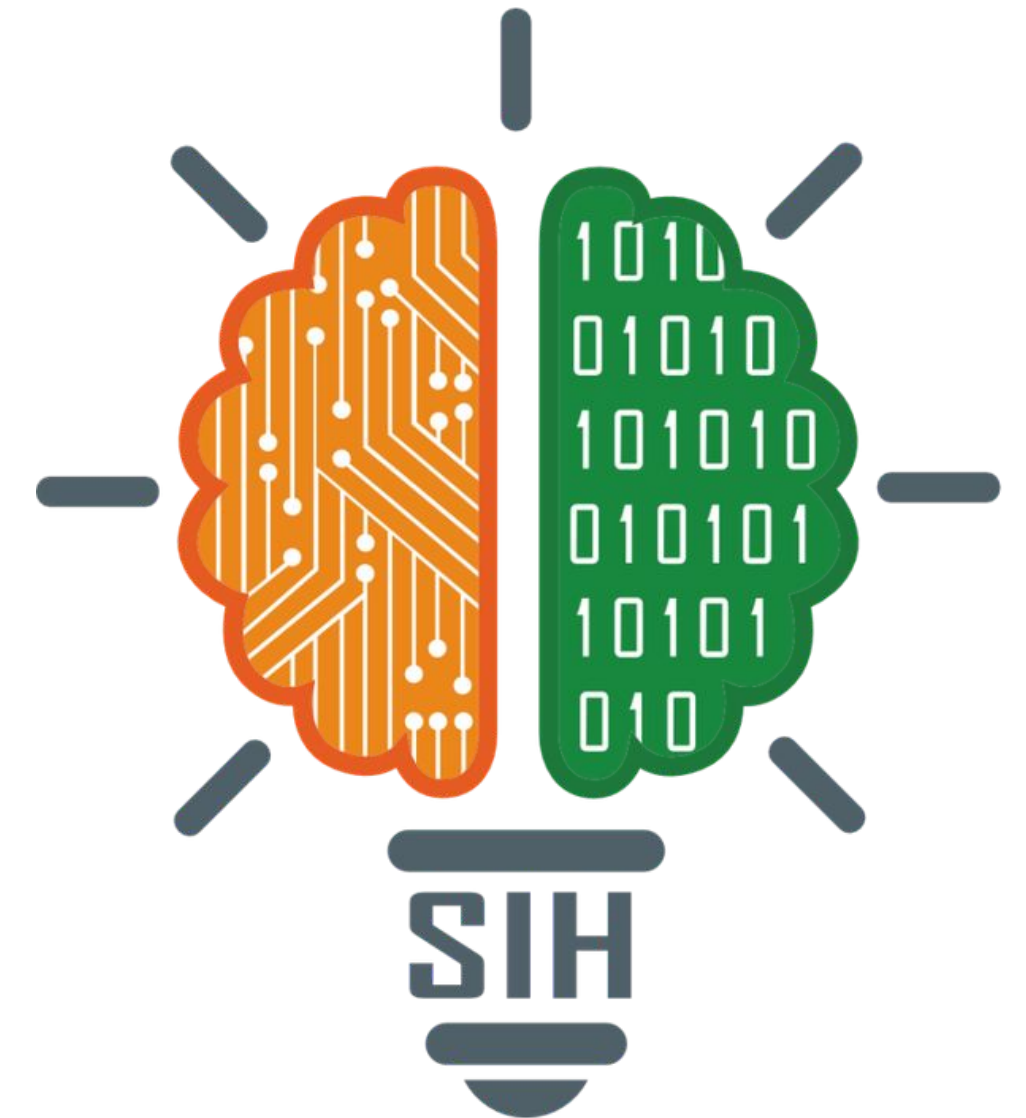


SMART INDIA HACKATHON 2026



TITLE PAGE

- Problem Statement ID – SIH26056
- Problem Statement Title – Real-Time Airfare Price Indexing for Consumer Price Index (CPI) Augmentation
- Team ID –
- Theme – Smart Automation
- PS Category – Software
- Team Name – 404 Found



- The Ministry of Statistics and Programme Implementation (MoSPI), via the National Statistical Office (NSO), measures and publishes India's monthly Consumer Price Index (CPI). This index dictates the country's inflation figures and heavily influences the Reserve Bank of India (RBI) when setting benchmark interest (repo) rates.

CORE INFLATION BLINDSPOT

Extreme Price Elasticity

A ticket bought 1 day before departure (T+1) can cost 300% to 500%
More than the exact same seat bought 30 days in advance (T+30).

Outdated Manual Point-in time Sampling

NSO traditionally collected static monthly quotes from physical airline offices, missing the full advance-purchase curve and passenger volume distributions.

Noise & Outlier Pollution

Flash sales, monsoon disruptions, and severe festival surges (Diwali/Chhath) heavily distort raw average calculations without robust statistical filtering.

Inter-Ministerial Silos

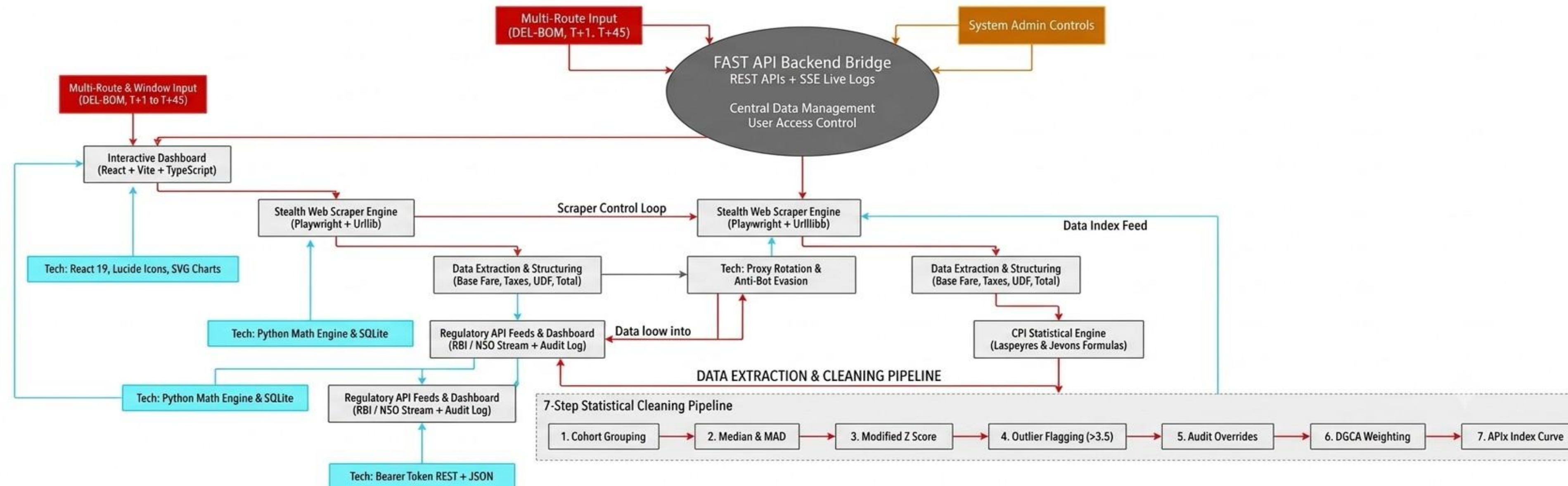
Downstream decision-makers (RBI Monetary Policy Committee, MoCA, DGCA) lack real-time, verifiable, automated statistical data feeds.

Tech Stack



- Backend: FastAPI (Python) on Port 3001 — REST API + SSE for real-time scraper log streaming; recalculates CPI after every scrape or override. Frontend: React + TypeScript + Vite — 5-module enterprise dashboard
- 3-layer architecture — mock portal server, standard HTTP scraper (rate-limited, proxy/UA rotation), and Playwright stealth scraper with anti-detection patches.
- Storage: SQLite (flights, index_history, overrides, settings) — auto-seeds 35+ days of historical data on startup.

Airline Fare Index System



Feasibility

Viability

Technical Feasibility

- Proven Stack: Built with lightweight, high-performance tech (FastAPI, React 19, SQLite/Postgres).
- Stealth Ingestion: Playwright headless scraping with proxy rotation bypasses bot blockers reliably.
- Zero-Downtime Design: Includes a mock portal fallback engine for offline testing and 100% demo stability.

2. Economic Feasibility

- Near-Zero Data Costs: Replaces expensive GDS API licenses (Amadeus/Sabre) with automated web extraction.
- Massive Cost Savings: Eliminates manual survey field visits, cutting operational costs by ~90%.
- Cloud-Ready: Lightweight containerized setup (Docker) deployable on NIC/MeghRaj government cloud.

3. Operational Feasibility

- Zero Coding Needed: Statisticians tune CPI weights & toggle formulas via intuitive UI sliders. Human-in-the-Loop: Outlier Audit Hub lets officers review & override flagged anomalies with mandatory audit notes.
- Plug-and-Play APIs: Delivers token-secured REST feeds directly to RBI & MoCA dashboards.

4. Legal & Ethical Feasibility

- Strict Compliance: Adheres to robots.txt and rate-limiting (10 req/min) to avoid target server load.
- Public Data Only: Scrapes only unauthenticated public fare quotes—zero PII or private data involved.

1. Long-Term Strategic Viability

- Government Alignment: Perfectly aligns with the Digital India and Data-Driven Governance mandates by modernizing NSO's 60-year-old manual CPI collection method.
- Critical National Utility: Directly empowers the Reserve Bank of India (RBI) Monetary Policy Committee with high-frequency inflation data to set accurate national interest (repo) rates.

2. Financial & Resource Sustainability

- Zero Recurring Licensing Fees: Avoids costly commercial aviation API subscriptions (Amadeus/IATA) by leveraging open-source stealth extraction.
- Low Maintenance Overhead: Automated pipeline requires minimal technical intervention, allowing small NSO IT teams to manage national-scale deployment effortlessly.

3. Market & Institutional Adoption

- High Inter-Ministerial Demand: Serves multiple government stakeholders simultaneously:
- MoSPI / NSO: Official CPI inflation reporting.
- MoCA & DGCA: Monitoring surge pricing during disasters or festival seasons.
- CCI (Competition Commission): Detecting potential airline cartelization on duopoly routes.

4. Scalability & Extensibility

- Nationwide Sector Expansion: Modular architecture seamlessly scales from 6 corridors to 1,500+ domestic routes, including subsidized UDAN regional flights.

Impact on Stakeholders

- MoSPI / NSO: automated real-time airfare data replaces manual monthly surveys, improving CPI accuracy.
- RBI Monetary Policy Committee: more precise transport-inflation input for repo-rate decisions affecting 1.4 billion people.
- DGCA & MoCA: real-time fare visibility supports consumer-protection enforcement and fare-policy design.
- Citizens: a more accurate inflation measure feeds into fairer interest rates on loans and deposits.

Benefits

- Structurally corrects a known blind spot in the CPI Services Basket (Transport & Communication sub-group).
- Full advance-purchase yield curve (T+1 to T+45) instead of one static monthly quote.
- Robust outlier governance — legitimate surges are kept, genuine anomalies are excluded.
- Complete audit-compliance trail builds institutional trust and transparency.
- Secure REST API makes the index directly consumable by RBI/NSO systems, closing inter-ministerial data silos.

➤ **MoSPI — Consumer Price Index Methodology**

Source: Ministry of Statistics & Programme Implementation — <https://www.mospi.gov.in>

➤ **IMF Consumer Price Index Manual (Jevons index guidance)**

Source: International Monetary Fund — <https://www.imf.org/en/Data/Statistics/cpi-manual>

➤ **DGCA Domestic Air Traffic & Fare Data**

Source: Directorate General of Civil Aviation — <https://www.dgca.gov.in>

➤ **RBI Monetary Policy Framework**

Source: Reserve Bank of India — <https://www.rbi.org.in>

➤ **FastAPI Documentation**

Backend framework reference — <https://fastapi.tiangolo.com>

➤ **Playwright Documentation**

Browser automation / scraping reference — <https://playwright.dev>